

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VIII

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Class : SY IT	Batch : 1st
Date of Assignment : 11-08-26	Date/Time of Submission : 11-08-26

Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

Code:

```
def bubbleSort(array):  
  
    for i in range(len(array)):  
  
        for j in range(0, len(array) - i - 1):  
  
            if array[j] > array[j + 1]:  
  
                temp = array[j]  
                array[j] = array[j+1]  
                array[j+1] = temp
```

```
data = [-2, 45, 0, 11, -9]
```

```
bubbleSort(data)
```

```
print('Sorted Array in Ascending Order:')  
print(data)  
print("Amit 44")
```

Output:

```
===== RESTART: C:/Users/11-47/App
```

```
Sorted Array in Ascending Order:
```

```
[-9, -2, 0, 11, 45]
```

```
Amit 44
```

```
Sorted Array in Ascending Order:
```

```
[-9, -2, 0, 11, 45]
```

```
Amit 44
```

b. Insertion sort

Code:

```
def insertionSort(array):
```

```
    for step in range(1, len(array)):
```

```
        key = array[step]
```

```
        j = step - 1
```

```
        while j >= 0 and key < array[j]:
```

```
            array[j + 1] = array[j]
```

```
            j = j - 1
```

```
    array[j + 1] = key
```

```
data = [9, 5, 1, 4, 3]
```

```
insertionSort(data)
```

```
print('Sorted Array in Ascending Order:')
```

```
print(data)
```

```
print("Amit 44")
```

Output:

```
===== RESTART: C:/Users/11-47/App
```

```
Sorted Array in Ascending Order:
```

```
[1, 3, 4, 5, 9]
```

```
Amit 44
```

```
Sorted Array in Descending Order:
```

```
[9, 5, 4, 3, 1]
```

```
Amit 44
```

c. Selection sort

Code:

```
def selectionSort(array, size):
```

```
    for step in range(size):
```

```
        min_idx = step
```

```
        for i in range(step + 1, size):
```

```
            if array[i] > array[min_idx]:
```

```
                min_idx = i
```

```
        (array[step], array[min_idx]) = (array[min_idx], array[step])
```

```
data = [-2, 45, 0, 11, -9]
```

```
size = len(data)
```

```
selectionSort(data, size)
```

```
print('Sorted Array in Descending Order:')
```

```
print(data)
```

```
print("Amit 44")
```

Output:

```
==== RESTART: C:/Users/IT-47/AI
```

```
Sorted Array in Descending Order:
```

```
[45, 11, 0, -2, -9]
```

```
Amit 44
```

```
----- RE
```

```
Sorted Array in Ascending Order:
```

```
[-9, -2, 0, 11, 45]
```

```
Amit 44
```

Curiosity Questions:

Programming Questions

1. Write a Python program to sort a list of numbers in ascending order using **Bubble Sort**.
2. Write a Python program to sort a list of numbers in ascending order using **Insertion Sort**.
3. Write a Python program to sort a list of numbers in ascending order using **Selection Sort**.

Theory Questions

4. What is sorting? Why is it required?
5. What is the basic idea of Bubble Sort?
6. How does Insertion Sort work?
7. How does Selection Sort work?
8. What is a **pass** in a sorting algorithm?
9. What is the difference between Bubble Sort and Selection Sort?
10. Which sorting algorithm is suitable when the data is already nearly sorted? Why?